DICE AND CUBES

Dice Positions, Cube Nets and Painted Cubes

OAVS TGT | Reasoning (Non-Verbal) | Topic 11: second non-verbal topic

Syllabus line (Reasoning, 30 Marks): “The Non-Verbal Reasoning topics like Figure Formation, **Dice**, Triangle, Rule Detection, Images, Completion of Pattern, Mirror Images, Figure Matrix and Paper Folding, etc.”

Built from: OAVS TGT English 2019, English 2022, Science 2024 and Social Studies 2023 (17 May 2023, Shift 2). Mathematics 2024 had none. All 5 dice and cube questions found in these papers are solved here with their original figures. Every PYQ answer and every practice answer was verified by a cube-folding / dice-orientation program.

What is inside

Ch.	Title	What you get
1	PYQ Analysis	Counts, patterns, priority, index of all 5 PYQs
2	Foundation	Cube facts, dice rules, the 11 cube nets, painted-cube formulas
3	Solved PYQs	Two dice questions, two net questions, one painted cube
4	Shortcuts, Traps and Scope	Speed methods and mistakes
5	Practice Set	13 original questions with a verified key
6	Quick Revision	One-page cheat sheet

CHAPTER 1

PYQ Analysis

Dice and cubes: 5 questions in four of the five papers

Paper	Q. numbers	Count	What was asked
English 2019	55	1	Net with triangular flaps: opposite of the face formed by the flaps
English 2022	40, 53	2	Painted cube cut into 64 cubes; net: face opposite 5
Mathematics 2024	none	0	—
Science 2024	96	1	Four positions of a dice: opposite of 2
Social Studies 2023	13	1	Three positions of a dice with letters: opposite of A
Total		**5**	**Found in 4 of the 5 papers**

Pattern	PYQs	Priority
Dice positions: find the opposite face from 3 or 4 views	2	★★★ Must
Cube net: find the opposite face (including a net with flaps)	2	★★★ Must
Painted cube cut into small cubes: count cubes by painted faces	1	★★ High
Expected, not yet asked: which cube can be made from a net, cuts needed, partly painted cubes	0	★★ Expected

PYQ INSIGHT: The dice in the PYQs are not ‘standard’ dice. In Science 2024 Q96 the opposite pairs come out as 1–6, 2–4 and 3–5 (not 7 in total). Never assume ‘opposite faces add up to 7’ unless the question says so.

PYQ INSIGHT: Both dice questions use the same idea: faces seen together are neighbours, so they cannot be opposite. Eliminate neighbours and one face remains.

Index of the 5 PYQs

Paper / Q	Type	Question in short	Ans	See
Eng 2019 Q55	Net with flaps	Cross-shaped net; opposite face of the face formed by the triangles	B	3.3
Eng 2022 Q40	Painted cube	Cube painted on opposite pairs; cut into 64; cubes with 3 painted faces	8	3.5
Eng 2022 Q53	Net	Net with 1 to 6; face opposite 5	2	3.4
Science 2024 Q96	Dice positions	Four views; opposite of 2	4	3.1
SST 2023 Q13	Dice positions	Three views of A to F; opposite of A	C	3.2

CHAPTER 2

Foundation

Cube facts, rules for dice, nets and painted cubes

2.1 Cube facts

Fact	Value
Faces / edges / corners	6 / 12 / 8
Faces meeting at a corner	3 (any two of them are neighbours)
Each face has	4 neighbours and exactly 1 opposite face
Two faces seen together in one view	are **neighbours**: they cannot be opposite

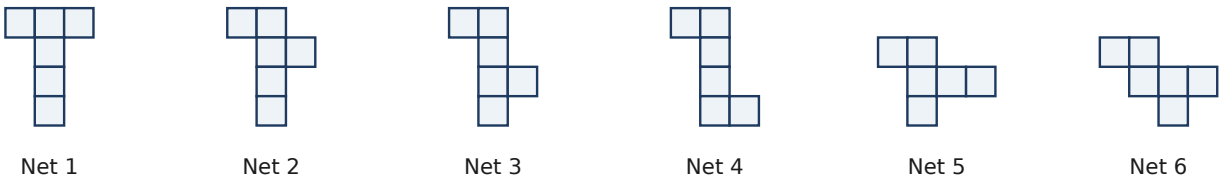
2.2 Dice positions: three tools

Tool	How it works	Used in
1. Neighbour elimination	List every face seen together with the target. Its opposite is a face never seen with it. If two candidates remain, use the face that is already opposite something else.	Science 2024 Q96
2. Same top face, rotate	If two views have the same face on top (or in the same place), the side faces go round in a fixed cyclic order. In a cycle of four, faces two apart are opposite.	SST 2023 Q13
3. Face never seen	A face that never appears is opposite one of the faces seen in every view, or opposite the face that has all other faces as neighbours.	Science 2024 Q96 (6 opposite 1)

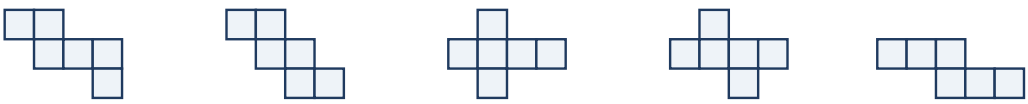
SHORTCUT: Never assume opposite faces add up to 7. Use the pictures. In Science 2024 Q96 the pairs are 1–6, 2–4, 3–5.

2.3 Cube nets

Exactly **11** flat shapes of six squares fold into a cube. Here they are, computed by folding each shape (the program also confirmed that no other of the 35 six-square shapes works):



Net 1 Net 2 Net 3 Net 4 Net 5 Net 6



Net 7 Net 8 Net 9 Net 10 Net 11

Net rule	Meaning	Example
Skip-one rule	In any straight line of three (or four) squares, the squares with exactly one square between them are **opposite**.	Eng 2022 Q53: 4 and 3, 1 and 6
Remaining pair	After using the rule twice, the two squares left are opposite.	2 and 5
Flaps (triangles)	Flaps on the outer edges of the four arms of a cross all meet on the face **opposite the centre**.	Eng 2019 Q55
Invalid nets	Four in a row with both extra squares on the **same** side (they overlap when folded), or any 2×2 block of squares.	—

2.4 Painted cubes

A big cube (n × n × n) has all 6 faces painted and is cut into n^3 small cubes. Count by position:

Painted faces on a small cube	Where	Formula	n = 3	n = 4	n = 5	n = 6
3	corners	8	8	8	8	8
2	edges (not corners)	12 (n − 2)	12	24	36	48
1	face centres	6 $(n-2)^2$	6	24	54	96
0	inside	$(n-2)^3$	1	8	27	64
Total		n^3	27	64	125	216

SHORTCUT: Cuts: to cut a cube into n^3 equal cubes you need **3 (n − 1)** cuts (n − 1 in each direction): 8 small cubes need 3 cuts, 27 need 6, 64 need 9. If only some faces are painted, count by layer: top-layer cubes have the top face painted, and so on.

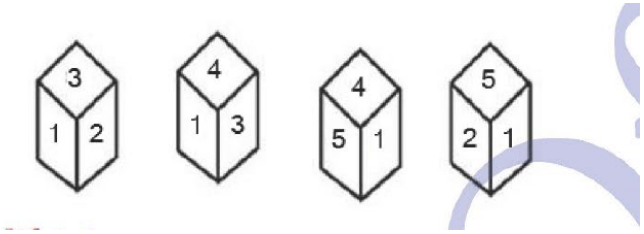
CHAPTER 3

Solved PYQs

All five questions with their original figures

3.1 Science 2024 Q96: four positions of a dice

Four different positions of the same dice are shown. Select the number that will be on the face opposite to the one having ‘2’. (1. 1 2. 3 3. 4 4. 5)

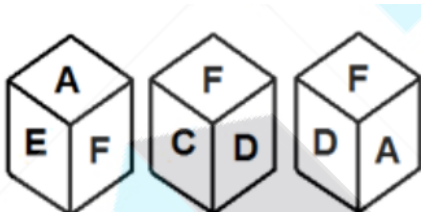


1. Faces seen together with 2: in view 1, 2 is with 3 and 1; in view 4, 2 is with 5 and 1. So 2 is next to 1, 3 and 5.
2. Face 1 is seen with 3, 2, 4 and 5 (all except 6), so the opposite of 1 must be 6, the face that never appears.
3. 2’s opposite is not 1, 3, 5 (neighbours) and not 6 (opposite of 1). Only 4 remains. Check: 4 is seen with 1, 3 and 5 and never with 2 ✓.
Answer: 4 (Option 3)

Face	Seen with (neighbours)	Opposite
1	2, 3, 4, 5	6
2	1, 3, 5	4
3	1, 2, 4	5

3.2 SST 2023 Q13: three positions with letters

Three different positions of the same dice are shown, the six faces labelled A to F. Select the letter on the face opposite to ‘A’. (1. F 2. C 3. D 4. B)

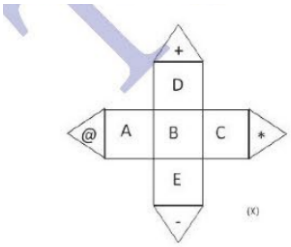


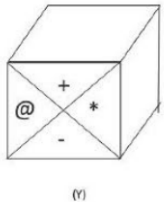
1. Views 2 and 3 have F on top. In view 2 the side faces are C (left) and D (right). In view 3 the side faces are D (left) and A (right).
2. Turning the dice around its vertical axis shifts C → D → A, so the four side faces go round in the order C, D, A, then a fourth face.
3. In a cycle of four, faces two apart are opposite: **C is opposite A**.
4. Check with view 1: A is next to E and F, never to C ✓.
Answer: C (Option 2)

> **SHORTCUT: Same face on top (or in the same position) in two views:** read the side faces as a cycle; the opposite of any face is two steps along the cycle.

3.3 English 2019 Q55: net with flaps

A paper as shown in figure (X) is folded into a cube as shown in figure (Y). Which option could represent the opposite face of the front face shown in (Y)? (A, B, D, C)

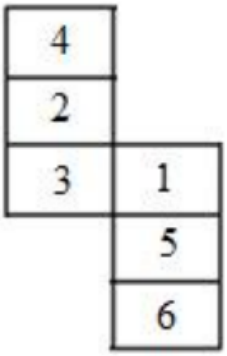




1. Figure X is a **cross**: B in the centre with A (left), C (right), D (up) and E (down). Triangular flaps @, *, + and − are attached to the outer edges of A, C, D and E.
2. Fold A, C, D and E up around B. Their outer edges, with the four flaps, meet to form the face **opposite B**.
3. Figure Y shows that face (the front face) divided into the four flaps @, +, *, −.
4. So the face opposite the front face is the centre square B.
Answer: B (Option B)

3.4 English 2022 Q53: net opposite of 5

A cube is made by folding the given sheet. What is the number on the face opposite the number 5? (A. 4 B. 6 C. 2 D. 3)



1. Left column 4, 2, 3: a straight line of three. The ends have exactly one square between them, so **4 and 3 are opposite**.
2. Right column 1, 5, 6: **1 and 6 are opposite**.
3. The remaining pair is 2 and 5: **2 is opposite 5**. (The folding program confirms: 4–3, 1–6, 2–5.)
Answer: 2 (Option C)

3.5 English 2022 Q40: painted cube

All the opposite faces of a big cube are painted yellow, green and purple respectively. The cube is cut into 64 small equal cubes. How many small cubes have three coloured faces? (A. 16 B. 24 C. 8 D. 12)

1. 64 small cubes means n = 4 (4 × 4 × 4). All six faces are painted (three colours, each on a pair of opposite faces).
2. A small cube has three painted faces only at a **corner** of the big cube. A cube has 8 corners.
3. Check with the table in 2.4: n = 4 gives 8 / 24 / 24 / 8 for 3 / 2 / 1 / 0 painted faces (total 64).
Answer: 8 (Option C)

> **TRAP:** 24 is the number with **two** painted faces (edges), and 16 or 12 come from mixing up the formulas. Three painted faces always mean the corners: 8.

CHAPTER 4

Shortcuts, Traps and Scope

Speed methods and mistakes

- **Dice:** two faces seen together are neighbours. Cross them off and the remaining face is the opposite.
- **Same face on top in two views:** read the other faces as a cycle; opposite = two steps along.
- **Nets:** in a straight line of three squares, the two end squares are opposite (skip-one rule).
- **Flaps on a cross:** all flaps meet on the face opposite the centre.
- **Painted cubes:** 3 faces = corners (8); 2 faces = edges 12 $(n-2)$; 1 face = 6 $(n-2)^2$; 0 = $(n-2)^3$.
- **Cuts:** 3 $(n-1)$ cuts for n^3 cubes.

Trap	How to avoid it
Assuming opposite faces add to 7	Only if the question says so. Use the views.
Using a face that is seen in no view as 'opposite' of a random face	A never-seen face is opposite a face that is next to all the others (here, 1).
Reading left and right of a dice view the wrong way	Compare views by the cyclic order, not by the picture alone.
Mixing painted-cube formulas	Use the table in 2.4: corners 8, edges 12 $(n-2)$, faces 6 $(n-2)^2$, inside $(n-2)^3$.
Not checking the net is valid	Four in a row with two squares on the same side is not a net (the program confirmed 11 valid nets).

Scope

PYQ	Where it belongs
Eng 2019 Q47, Q48, Q49; Eng 2022 Q31	Paper Folding (folding and cutting, transparent sheet): a separate PDF
Eng 2022 Q32	Counting figures: Triangle topic

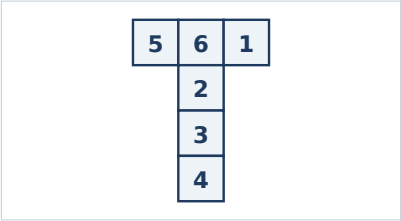
CHAPTER 5

Practice Set

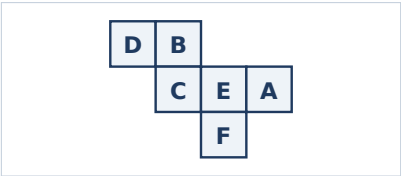
13 original questions; answers on the next page (all verified by program)

Part A: Cube nets. Which face is opposite?

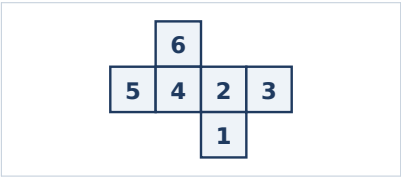
Q1. The net below is folded into a cube. Which face is opposite to **3**?



Q2. The net below is folded into a cube. Which face is opposite to **B**?

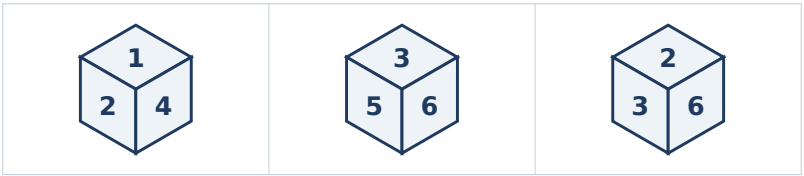


Q3. The net below is folded into a cube. Which face is opposite to **1**?

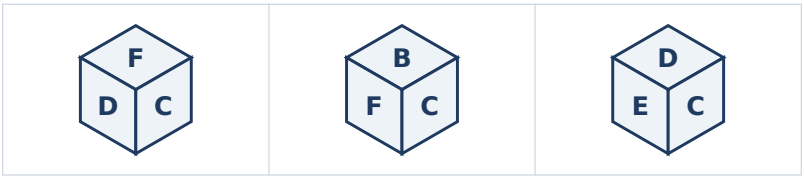


Part B: Dice positions. Which face is opposite?

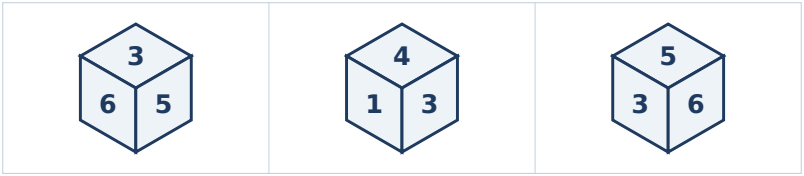
Q4. Three positions of the same dice are shown. What is on the face opposite **1**?



Q5. Three positions of the same dice are shown. What is on the face opposite **B**?



Q6. Three positions of the same dice are shown. What is on the face opposite **1**?



Part C: Painted cubes and cuts

Q7. A 4 × 4 × 4 cube, painted red on all faces, is cut into 64 small cubes. How many small cubes have exactly **two** faces painted?

Q8. A 5 × 5 × 5 cube, painted on all faces, is cut into 125 small cubes. How many have exactly **one** face painted?

Q9. A 3 × 3 × 3 cube, painted on all faces, is cut into 27 small cubes. How many have **no** face painted?

Q10. A 6 × 6 × 6 cube, painted on all faces, is cut into 216 small cubes. How many have exactly **two** faces painted?

Q11. A 4 × 4 × 4 cube has only its **top and bottom** faces painted. It is cut into 64 small cubes. How many have exactly one painted face?

Q12. How many cuts are needed to cut a cube into 64 equal small cubes?

Q13. A 3 × 3 × 3 painted cube is cut into 27 small cubes. How many have **at least one** painted face?

Answer key

Q	Answer	Reason
1	6	Net folded by program: 3 is opposite 6. Use the skip-one rule on a straight line of three.
2	F	Net folded by program: B is opposite F. Use the skip-one rule on a straight line of three.
3	6	Net folded by program: 1 is opposite 6. Use the skip-one rule on a straight line of three.
4	6	All consistent dice labellings give 6 opposite 1: faces seen with 1 cannot be opposite.
5	D	All consistent dice labellings give D opposite B: faces seen with B cannot be opposite.
6	5	All consistent dice labellings give 5 opposite 1: faces seen with 1 cannot be opposite.
7	24	Edges: $12 \times (4 - 2) = 24$
8	54	Face centres: $6 \times (5 - 2)^2 = 6 \times 9 = 54$
9	1	Inside: $(3 - 2)^3 = 1$
10	48	Edges: $12 \times (6 - 2) = 48$
11	32	Only the top and bottom layers have a painted face: 16 + 16 = 32 (each of those cubes has exactly one painted face)
12	9	$3 \times (4 - 1) = 9$ cuts (3 in each direction)
13	26	$27 - 1$ (the centre cube) = 26

CHAPTER 6

Quick Revision

One page for the last day

Question	Method	PYQ check
Dice positions	Faces seen together are neighbours; remove them; use the face already opposite another	Sci 2024 Q96: 2 opposite 4
Same top face in two views	Cycle of side faces; opposite = two steps	SST 2023 Q13: A opposite C
Net with 1-2-3 lines	Skip-one rule: ends of a line of three are opposite	Eng 2022 Q53: 5 opposite 2
Cross net with flaps	Flaps meet on the face opposite the centre	Eng 2019 Q55: B
Painted cube cut up	Corners 8; edges 12 $(n-2)$; faces 6 $(n-2)^2$; inside $(n-2)^3$	Eng 2022 Q40: 8

ROUTINE: Exam-day routine: write which faces are neighbours; cross them off; one face is left. For nets, find the straight lines of three. 40 seconds per question. Next in the syllabus order: Triangle, Rule Detection, Images, Completion of Pattern, Mirror Images, Figure Matrix, Paper Folding.